

Azure Networking — Zero to Practical

Explained through the "Secure Housing Society" analogy

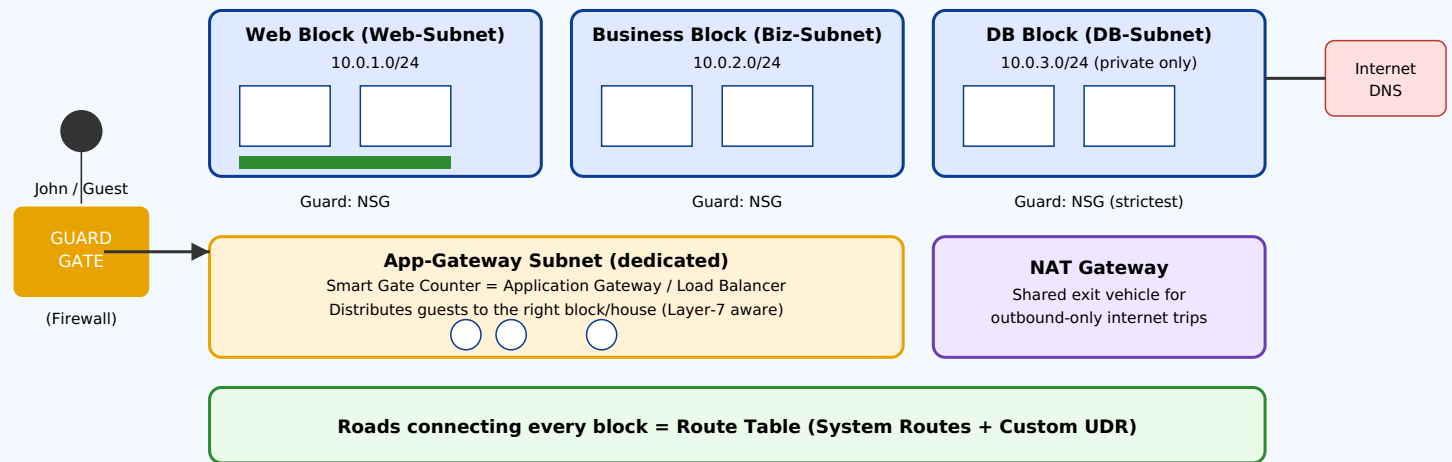
Core idea: Every Azure networking component maps 1:1 to something in a gated housing society. If you remember the society, you remember Azure networking.

Covers: VNET, CIDR, Subnets, NSG, ASG, Route Tables (System + UDR), NAT Gateway, Load Balancer, Application Gateway, Azure Firewall, DNS, VNET Peering, VNET Gateway, VPN Gateway, Resource Hierarchy.

Format: Crisp notes + diagrams + comparison tables. Built for revision, not bedtime reading.

1. The Big Picture — One Diagram to Rule Them All

VNET = "Secure Housing Society" (10.0.0.0/16)



Read it like this: John (a resident) or a Guest arrives at the Guard Gate (Firewall). Once inside, the society (VNET) is divided into Blocks (Subnets) — Web, Business, DB — each with its own guard (NSG). A Guard Gate counter with multiple lanes (App Gateway / Load Balancer) directs guests to the right block. Roads (Route Table) connect every block. A shared exit vehicle (NAT Gateway) lets residents go out to the city without exposing their home address.

2. VNET & CIDR — The Land Itself

Housing analogy: The entire plot of private land the society is built on. It has a boundary wall and one official address range printed on the society's registration papers.

Azure: A **Virtual Network (VNET)** is an isolated private network inside a region (e.g. East-US). Its size/address range is defined by a **CIDR block**, e.g. $10.0.0.0/16 = 65,536$ private IPs available to carve up.

- VNET = logical boundary. Nothing outside can see inside unless you explicitly allow it (peering, VPN, public IP, gateway).
- CIDR = how big the land is and how it's numbered. Bigger prefix number (/24) = smaller land; smaller number (/16) = bigger land.
- A VNET lives in one region. Two VNETs in two regions = two separate societies in two separate cities.

Common confusion: VNET is NOT a physical network — it's a logical isolation boundary Azure enforces in software. Think "gated society on paper," not "actual bricks."

3. Subnets — Blocks Inside the Land

Housing analogy: The land is divided into Block 1, Block 2, Block 3 — Web Block, Business Block, DB Block. Each block gets its own slice of the total address range.

Azure: A **Subnet** is a sub-range of the VNET's CIDR, carved out for a specific tier/purpose. E.g. VNET 10.0.0.0/16 → Web-Subnet 10.0.1.0/24, Business-Subnet 10.0.2.0/24, DB-Subnet 10.0.3.0/24.

Block (Society)	Subnet (Azure)	Typical purpose
Guest parking / entrance block	App-Gateway Subnet	Holds Application Gateway / Load Balancer only
Residential block A	Web-Subnet	Web servers / front-end VMs
Residential block B	Business-Subnet	App/business logic servers
Locked-down block (no outsiders)	DB-Subnet	Databases — no direct internet exposure

Why split into subnets at all? Same reason a society doesn't let guests wander everywhere — segmentation lets you apply different security rules and routing per tier. DB block gets the tightest guard rules; Web block is more open (has to talk to internet).

4. NSG (Network Security Group) — The Guard Per Block

Web-Subnet



= NSG

Rule: Allow port 443
from Internet

Rule: Deny all else

(checks every visitor at block entry)

DB-Subnet



= NSG

Rule: Allow port 1433
only from Biz-Subnet

Rule: Deny Internet

(much stricter guard)

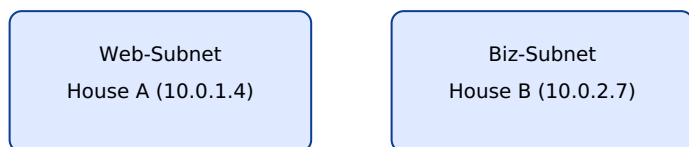
Housing analogy: Every block has its own security guard who checks ID before letting anyone in/out of that specific block. The Web block's guard is relaxed (has to let visitors in). The DB block's guard is paranoid — only lets in people already verified by the Business block.

Azure: NSG = a stateful firewall attached to a subnet (or a NIC) with allow/deny rules based on source/destination IP, port, protocol. Rules are evaluated by priority number, lowest first.

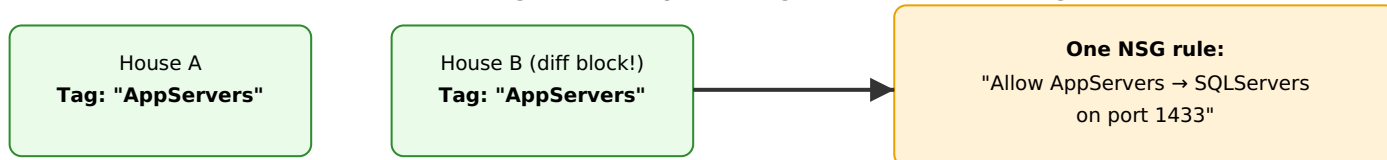
NSG can attach at two levels: subnet level (guard at block entrance — applies to everyone in the block) or NIC level (guard at the individual house door — applies to one VM only). You can use both together.

5. ASG (Application Security Group) — Guards by Role, Not Location

Without ASG: write one NSG rule per house/IP (messy)



With ASG: tag houses by ROLE, guard checks the tag



Housing analogy: Instead of the guard remembering every house number, every resident wears a badge saying their role — "Doctor," "Delivery Staff," "Owner." The guard's rule is "Delivery Staff allowed only 9am-9pm" — works no matter which block that person's house is in.

Azure: ASG groups VMs by application role (e.g. "WebTier," "AppTier," "SqlTier") regardless of which subnet/IP they're in. You write NSG rules against ASG names instead of hardcoded IPs — rules survive VM scaling/IP changes.

NSG vs ASG in one line: NSG = the guard + rulebook. ASG = the badge that groups who the rule applies to.

6. Route Table — Roads Connecting the Blocks

Housing analogy: Roads inside the society are auto-built the moment blocks are created (default roads = every block connects to every other block + main gate). If management wants traffic to compulsorily pass through a checkpoint/clubhouse first, they lay a custom road overriding the default.

Azure:

- **System Routes** — created automatically: subnet-to-subnet, VNET-to-internet, VNET-to-peered-VNET. You never configure these; they just exist.
- **User Defined Routes (UDR / custom Route Table)** — you override the default road, e.g. force all outbound traffic to go through a Firewall or NVA (Network Virtual Appliance) before leaving.

Exam trap: Route table controls WHERE traffic goes (path). NSG controls WHETHER traffic is allowed (permission). Both can exist independently — traffic can have a valid road (route) but still get blocked at the guard (NSG).

7. NAT Gateway — Shared Exit Vehicle

Housing analogy: Residents inside private blocks don't each drive their own registered car out to the city (would expose each house's private details). Instead, they use one shared society vehicle at the exit with one shared number plate. Outsiders can never use that plate to drive back in.

Azure: NAT Gateway gives outbound-only internet access to resources in a subnet using one (or a few) shared public IP(s), without giving those resources their own public IP or exposing them to inbound internet traffic.

NAT Gateway vs Load Balancer outbound rules: both provide outbound connectivity, but NAT Gateway is simpler, scales automatically, and is the modern recommended approach for pure outbound-only traffic.

8. Load Balancer vs Application Gateway — The Smart Gate Counter

Housing analogy	Load Balancer	Application Gateway
What it does at the gate	Sends each arriving car to the next free counter, round-robin — doesn't care where the car is going	Reads the visitor's slip ("visiting House 12 for Plumbing") and routes them to the exact right block/queue
OSI Layer	Layer 4 (Transport — IP + port)	Layer 7 (Application — reads URL path, host header)
Best for	Fast, simple distribution across identical VMs	URL-based routing, SSL termination, WAF (Web Application Firewall) built in
Scope	Regional or global (Azure LB)	Regional, needs its own dedicated subnet

From your screenshots: the "App-Gateway Subnet" needs to be its own dedicated subnet — Application Gateway cannot share a subnet with VMs. This matches "3.4.6.6," "LB," "abhishek.com" routing shown in the whiteboard — the App Gateway/LB terminates the incoming request and forwards to the right backend pool (Web-Subnet VMs).

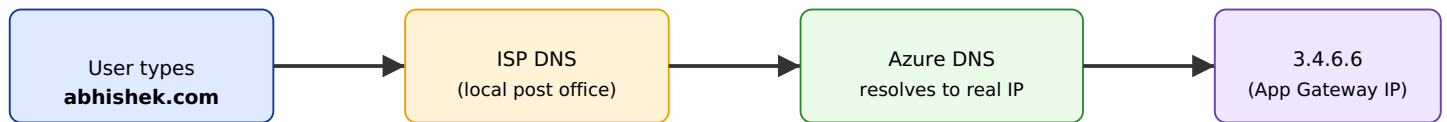
9. Azure Firewall — The Central Boundary Checkpoint

Housing analogy: The main boundary wall + the single main gate checkpoint for the ENTIRE society — not per-block. Every vehicle in or out of the society (regardless of which block it's headed to) passes this one checkpoint first.

Azure: Azure Firewall is a centralized, managed network firewall for the whole VNET (or hub in hub-spoke design). It filters traffic by FQDN, IP, port, protocol — applied once, protects everything behind it.

	NSG	Azure Firewall
Scope	Per subnet/NIC (local guard)	Whole VNET / hub (central checkpoint)
Filtering	IP/port/protocol only	IP/port/protocol + FQDN + threat intelligence
Analogy	Block-level guard	Main society gate + wall

10. DNS — Society Name to Actual Address

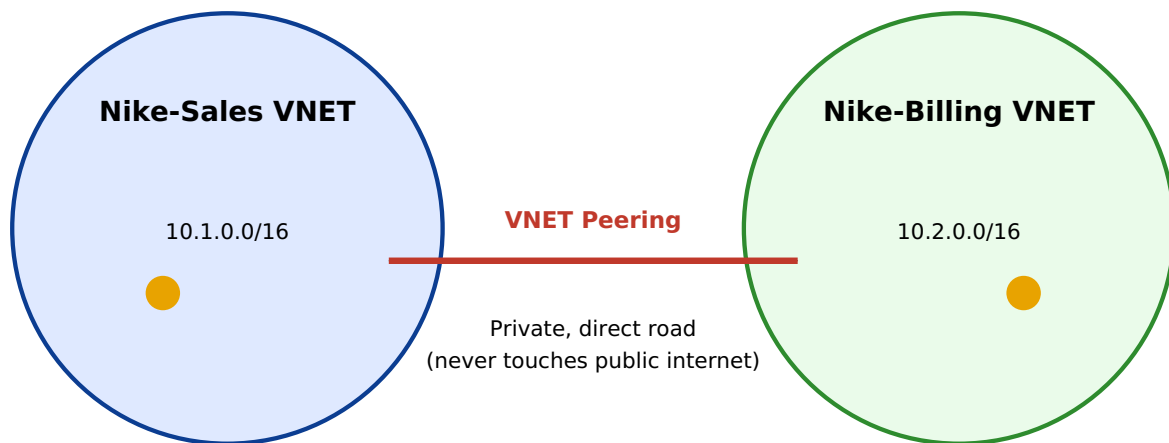


Same as asking a stranger "where's Secure Housing Society?" → post office looks it up → gives GPS coordinates

Housing analogy: "Secure Housing Society" is a friendly name. Nobody navigates by GPS coordinates directly — they ask around (ISP/local directory), which points to the actual society location.

Azure: DNS translates a friendly name (abhishek.com) into an IP address. **Azure DNS** hosts your domain's records; public resolution (like google.com, amazon.com in your notes) works the same way — hierarchical lookup through ISP DNS → authoritative DNS.

11. VNET Peering — Two Societies, One Private Road



Housing analogy: Two independent societies right next to each other (Nike-Sales, Nike-Billing) build a private direct road between them so residents can visit each other without going onto the public highway. Each society keeps its own management, guards, rules — they're just now connected.

Azure: VNET Peering connects two VNETs (same or different regions — "Global Peering") using Microsoft's private backbone, low latency, no public internet hop. VNETs stay isolated/independently managed by default — peering just enables routing between them.

Key limitation: Peering is NOT transitive. If A peers with B, and B peers with C, A cannot automatically reach C. You'd need A↔C peered directly, or use a hub-spoke model with a central hub VNET.

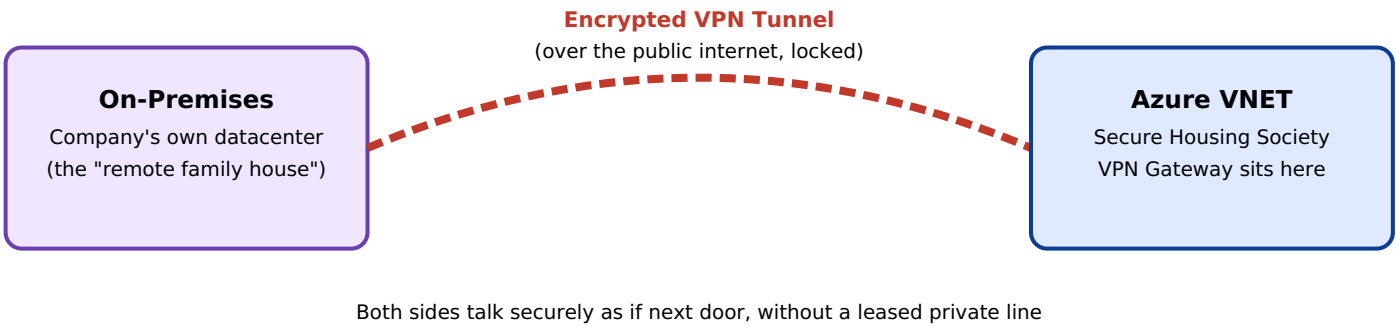
12. VNET Gateway — The Dedicated Connector Checkpoint

Housing analogy: For peering, two societies just build a direct road — no special toll booth needed. But for anything going OUTSIDE the private road system (e.g. a tunnel to a remote family house, or a VPN-style secure link), you need a dedicated checkpoint/toll-booth structure — that's the gateway.

Azure: A **Virtual Network Gateway** is a special gateway resource deployed into its own "GatewaySubnet" inside the VNET. It's the resource that actually terminates VPN or ExpressRoute connections. Two types: **VPN Gateway** (site-to-site/point-to-site over internet, encrypted) and **ExpressRoute Gateway** (private dedicated circuit, not over internet).

Note: Peering does NOT need a VNET Gateway. Gateways are needed specifically for VPN / ExpressRoute connections — this is a common exam mix-up.

13. VPN Gateway — Tunnel to the Remote Family House (On-Premises)



Housing analogy: One family member still lives at the old family house (on-premises datacenter) outside the society. Instead of exposing the whole city road to reach them, the society builds a locked, dedicated tunnel under the public road — only that family can use it, and everything sent through it is sealed.

Azure: VPN Gateway (site-to-site) creates an encrypted IPsec tunnel over the public internet between an on-premises network and a VNET. Lets on-prem resources and Azure resources communicate as if on the same network. Cheaper than ExpressRoute but rides on public internet (encrypted, still not a dedicated private line).

VPN Gateway		ExpressRoute
Path	Over public internet (encrypted)	Private dedicated circuit (not internet)
Setup speed	Fast, cheap	Slower to provision, pricier
Use case	Small/medium hybrid setups, quick connect	Large enterprise, needs guaranteed bandwidth/low latency

14. Governance Hierarchy — Who Manages/Bills What

Housing analogy: City → Township → Society → Block → House. Each level has its own management/billing office, but policy set at the City level cascades down to every house.

Azure: Management Group → Subscription → Resource Group → Resource. "Azure Dev / Mgmt / Billing" in your notes = this exact hierarchy — subscriptions are billing boundaries, resource groups are logical containers for related resources (like one Society's block registry).

15. Master Cheat-Sheet — Housing Society ↔ Azure Networking

Housing Society	Azure Component
Private land with boundary wall	VNET
Land's registered address range	CIDR block
Blocks inside the land	Subnets
Security guard per block	NSG
Badge grouping residents by role, not block	ASG
Roads connecting blocks	Route Table (System Routes + UDR)
Shared exit vehicle for outbound-only trips	NAT Gateway
Multi-counter gate distributing guests evenly	Load Balancer (Layer 4)
Smart gate reading visitor's exact destination	Application Gateway (Layer 7) + WAF
Main boundary wall + central checkpoint	Azure Firewall
Society name resolved to GPS coordinates	DNS (Azure DNS / public DNS)
Private direct road between two societies	VNET Peering
Dedicated toll-booth for tunnels/VPN	Virtual Network Gateway
Locked tunnel to a remote family house (on-prem)	VPN Gateway
Private leased road (no public street at all)	ExpressRoute
City → Township → Society → Block → House	Mgmt Group → Subscription → Resource Group → Resource

TRAFFIC FLOW RECAP: User → DNS (name→IP) → Firewall/Guard Gate → App Gateway/LB (picks right block) → Subnet NSG (guard check) → VM/House → (if talking to on-prem) → VPN Gateway tunnel → On-premises → (if talking to another VNET) → VNET Peering → Other VNET